

**Aim:** To study the variation in cell potential of the cell  $\text{Zn}|\text{Zn}^{2+}||\text{Cu}^{2+}|\text{Cu}$  with change in concentration of electrolytes ( $\text{CuSO}_4/\text{ZnSO}_4$ ) at room temperature.

**Theory:** The cell under investigation in this experiment is represented as follows:



Here  $x\text{M}$  denotes varying concentrations of  $\text{Cu}^{2+}(\text{aq})$  ions. In other words, to study the variation in cell potential with concentration, the concentration of  $\text{Cu}^{2+}(\text{aq})$  is varied while that of  $\text{Zn}^{2+}(\text{aq})$  is kept constant. The measured cell potential enables us to calculate the electrode potential of  $\text{Cu}^{2+}/\text{Cu}$  electrode for each concentration of copper (II) ions. This variation is theoretically depicted according to the equation :

$$E_{\text{Cu}^{2+}/\text{Cu}} = E_{\text{Cu}^{2+}/\text{Cu}}^0 + \frac{0.05916}{2} \log [\text{Cu}^{2+}] \dots\dots\dots \text{I}$$

The variation in the electrode potential of  $\text{Cu}^{2+}/\text{Cu}$  electrode consequently brings variation in the cell potential according to the relation :

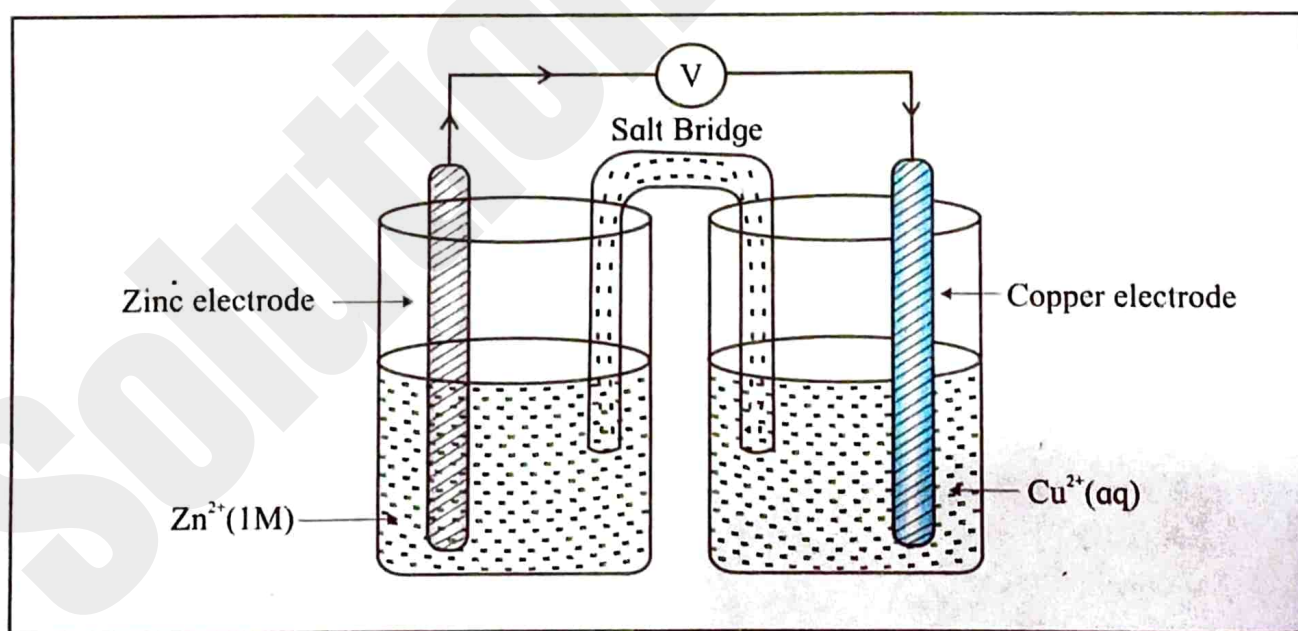
$$E_{\text{cell}} = E_{\text{Cu}^{2+}/\text{Cu}} - E_{\text{Cu}^{2+}/\text{Cu}}^0 \dots\dots\dots \text{II}$$

Equation (II) clearly suggests that even if  $E_{\text{Zn}^{2+}/\text{Zn}}^0$  is kept constant, the variation in  $E_{\text{Cu}^{2+}/\text{Cu}}$  would bring corresponding variation in  $E_{\text{cell}}$  (cell potential). Similarly, keeping the concentration of  $\text{Cu}^{2+}$  ions constant, one can study the variations in the cell potential with the variation in concentration of  $\text{Zn}^{2+}$  ions.

**Apparatus:** Zinc plate (or rod), copper plate (or rod), beakers (50 cm<sup>3</sup>), voltmeter (potentiometer), salt bridge, etc.

**Chemicals:** 1.0 M  $\text{ZnSO}_4$  solution, 0.2 M, 0.1 M, 0.05 M, 0.025 M and 0.0125 M  $\text{CuSO}_4$  solutions, etc.

**Diagram:**



### Procedure:

1. Set up the cell as given in diagram, using 1.0 M  $\text{ZnSO}_4$  and 0.2 M  $\text{CuSO}_4$  solutions.
2. Measure the potential difference of the cell and also keep record of the polarity of the electrodes (this will enable us to give a sign to the cell potential  $E_{\text{cell}}$ ).
3. Remove the salt bridge as soon as the cell potential measurement is over.
4. Replace the beaker of 0.2 M  $\text{CuSO}_4$  with 0.1 M  $\text{CuSO}_4$  solution in the beaker. Place the salt bridge in position and note the cell potential.
5. Repeat this procedure for other solutions of copper sulphate in decreasing order of concentrations of copper sulphate solution.
6. Calculate  $\log_{10} [\text{Cu}^{2+}(\text{aq})]$  and then  $E_{\text{Cu}^{2+}/\text{Cu}}$  for each variation in the concentration of copper (II) solution.
7. Record electrode potential values of  $\text{Cu}^{2+}(\text{aq})/\text{Cu}(\text{s})$  electrode for different concentrations of  $\text{Cu}^{2+}$  ions as given observation table.
8. Plot a graph for the  $\log_{10} [\text{Cu}^{2+}]$  against  $E_{\text{cell}}$ .

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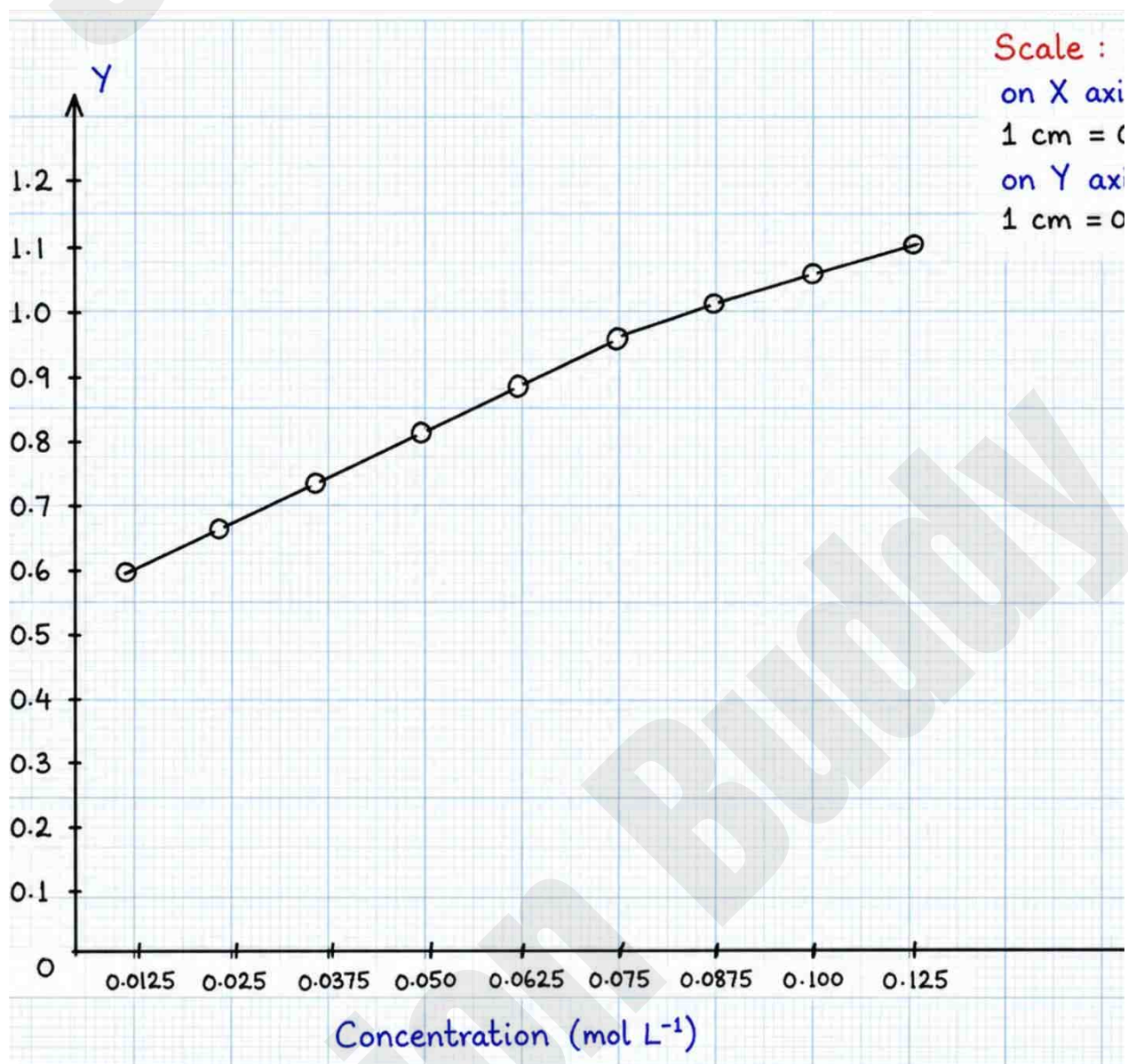
### Observation Table :

| Sr. No. | $[\text{Cu}^{2+}(\text{aq})]$ in mol/L | $\log [\text{Cu}^{2+}(\text{aq})]$ in mol/L | $E_{\text{cell}}$ in V | $E(\text{Cu}^{2+}/\text{Cu})$ Experimental value in V |
|---------|----------------------------------------|---------------------------------------------|------------------------|-------------------------------------------------------|
| 1.      | 0.2                                    | 1.3010                                      | 1.068                  | 0.298                                                 |
| 2.      | 0.1                                    | 1.0000                                      | 1.062                  | 0.292                                                 |
| 3.      | 0.05                                   | 2.6990                                      | 1.054                  | 0.284                                                 |
| 4.      | 0.025                                  | 2.3679                                      | 1.044                  | 0.274                                                 |
| 5.      | 0.0125                                 | 2.0969                                      | 1.033                  | 0.263                                                 |

### Result/Conclusion:

The variation in cell potential with change in concentration of electrolyte at room temperature is

EMF of the cell decrease with decrease in concentration of  $\text{Cu}^{2+}$  ions.





**MCQ**

Select [✓] the most appropriate answer from given alternatives of each sub question.

1. The potential difference between two electrodes of a galvanic cell is called.....
- ✓ a. cell potential                      b. standard oxidation potential  
c. standard reduction potential        d. half cell potential

$$2. \quad E_{M^{n+}/M} = E^0_{M^{n+}/M} - \frac{RT}{nF} \ln \frac{[M]}{[M^{n+}]}$$

This equation is called .....

- a. Einstein's equation ✓ b. Nernst's equation  
c. Boyle's equation d. Charle's equation
3. In Daniell cell, the cell potential is always.....  
a. negative b. may be negative  
c. may be positive ✓ d. positive

4. In following set up of cell,

Zn(s)|Zn<sup>2+</sup> (1.0 M)||Cu<sup>2+</sup> (xM)|Cu(s) anode is.....

- a. copper electrode                      b.  $\text{Cu}^{2+}$   
c.  $\text{Zn}^{2+}$                                       ✓ d. zinc electrode
5. What will be the nature of a graph of  $\log_{10} [\text{Cu}^{2+}(\text{aq})]$  against cell potential
- a. straight line passing through origin  
✓ b. straight line intersecting 'y' axis  
c. first increasing and then remains constant  
d. intersecting 'x' axis with decreasing slope.

### Short answer questions

1. It is necessary to clean the zinc and copper strips or rods with sand paper. Why?

Ans. Yes, it necessary to clean the zinc in the copper strip for the rod with Sandpaper because they don't correct it.

2. What will happen if we connect copper strip to negative terminal and zinc strip to positive terminal of voltmeter?

Ans. EMF of cell increases with decrease in the concentration of the electrolyte around the anode & increase in the concentration of the electrolyte around the cathode.

3. Which chemicals are used for preparation of salt bridge?

Ans. one type of salt bridge consists of a unshape glass tube filled with a nealative Li inert electrolytes.

4. Explain the role of salt bridge in this experiment?

A salt bridge is a connection of weak electrolyte between the oxidation and reduction half-cells in a galvanic cell. Its purpose is to maintain electrical neutrality by allowing ions to flow between the half-cells and to prevent the electrochemical reaction from reaching equilibrium too quickly.

Remark and sign of teacher: